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BBO on interval (BBO)

BBO on interval (BBO) provides the last best bid, best offer, and sale at 1-second or 1-minute intervals. This is a subset of [MBP-1](#).

Databento groups this under [top of book \(L1\)](#).

Databento adopts the following convention for BBO:

- The time interval is indicated by the schema ID's suffix: `-1s` for 1-second and `-1m` for 1-minute.
- If no trade or BBO update occurs within the interval, no record is printed.
- If a BBO update occurs but no trade takes place within the interval, the last sale information is forward-filled from the previous interval.
- If no BBO update occurs but a trade takes place within the interval, the BBO information is forward-filled from the previous interval.

Fields (`bbo-1s` and `bbo-1m`)

Field	Type	Description
<code>ts_recv</code>	<code>uint64_t</code>	The end timestamp of the interval, clamped to the second/minute boundary, expressed as the number of nanoseconds since the UNIX epoch.
<code>ts_event</code>	<code>uint64_t</code>	The matching-engine-received timestamp of the last trade for the instrument expressed as the number of nanoseconds since the UNIX epoch. Will be <code>UNDEF_TIMESTAMP</code> in cases where there was no last trade in the session. See ts_event .
<code>rtype</code>	<code>uint8_t</code>	A sentinel value indicating the record type. Can be either 195 (<code>BBO-1s</code>) or 196 (<code>BBO-1m</code>). See Rtype .
<code>publisher_id</code>	<code>uint16_t</code>	The publisher ID assigned by Databento, which denotes the dataset and venue. See Publishers .
<code>instrument_id</code>	<code>uint32_t</code>	The numeric instrument ID. See Instrument identifiers .
<code>side</code>	<code>char</code>	The side that initiated the last trade. Can be Ask for a sell aggressor in a trade, Bid for a buy aggressor in a trade, or None where no side is specified. See Side .
<code>price</code>	<code>int64_t</code>	The last trade price where every 1 unit corresponds to 1e-9, i.e. 1/1,000,000,000 or 0.000000001. Will be <code>UNDEF_PRICE</code> if there was no last trade in the session. See Prices .
<code>size</code>	<code>uint32_t</code>	The last trade quantity. Will be 0 if there was no last trade in the session.
<code>flags</code>	<code>uint8_t</code>	A bit field indicating event end, message characteristics, and data quality. See Flags .
<code>sequence</code>	<code>uint32_t</code>	The message sequence number assigned at the venue of the last update.
<code>bid_px_00</code>	<code>int64_t</code>	The bid price at the top level where every 1 unit corresponds to 1e-9, i.e. 1/1,000,000,000 or 0.000000001. See Prices .
<code>ask_px_00</code>	<code>int64_t</code>	The ask price at the top level where every 1 unit corresponds to 1e-9, i.e. 1/1,000,000,000 or 0.000000001. See Prices .
<code>bid_sz_00</code>	<code>uint32_t</code>	The bid size at the top level.
<code>ask_sz_00</code>	<code>uint32_t</code>	The ask size at the top level.
<code>bid_ct_00</code>	<code>uint32_t</code>	The bid order count at the top level.
<code>ask_ct_00</code>	<code>uint32_t</code>	The ask order count at the top level.

Consolidated BBO on interval (CBBO)

Consolidated BBO on interval (CBBO) provides the consolidated last best bid, best offer, and sale at 1-second or 1-minute intervals. This is a subset of [CMBP-1](#).

Databento adopts the following convention for CBBO:

- The time interval is indicated by the schema ID's suffix: `-1s` for 1-second and `-1m` for 1-minute.
- If no trade or CBBO update occurs within the interval, no record is printed.
- If a CBBO update occurs but no trade takes place within the interval, the last sale information is forward-filled from the previous interval.
- If no CBBO update occurs but a trade takes place within the interval, the CBBO information is forward-filled from the previous interval.

Fields (`cbbo-1s` and `cbbo-1m`)